EC-402: Impact Evaluation Methods

Delhi School of Economics | Winter Semester 2025–26

Assignment 1: Mosquito Nets and Dengue Risk

Posted: [07—03-2026] | ***Submission: [22-03-2026], 12 noon*** *via Google Form (TB posted)*

Background

You are a researcher investigating whether using mosquito nets reduces a household's risk of contracting dengue fever. You have been given a dataset of 1,752 households from an undisclosed district in India. The data is **observational** — households made their own decisions about whether to use nets, and you, as a researcher, have no control over this.

The district has a government programme that provides free mosquito nets to households meeting **both** of the following eligibility criteria:

- Household size strictly greater than 4 members, AND
- Monthly household income strictly below Rs. 7,000

Eligible households were **not** automatically enrolled — many chose not to participate. Non-eligible households could still purchase and use their own nets.

The dataset contains the following variables:

Variable	Type	Description
id	Integer	Household identifier
net	Binary	1 if household uses at least one mosquito net; 0 otherwise
net_num	Count	Number of mosquito nets used (0, 1, 2, or 3)
dengue_risk	Integer	Risk of dengue infection (0–100; higher = greater risk). This is your outcome variable.
income	Integer	Household monthly income — Note: multiply by 10 to obtain the value in Indian Rupees
health	Integer	Self-reported household health (0–100; higher = better health)
household	Integer	Number of members in the household
temperature	Float	Average nighttime temperature (°C)
resistance	Integer	Mosquito insecticide resistance index (0–100; higher = more resistant)
rainfall	Integer	Average monthly rainfall (mm)

How to Write Your Answer

This assignment asks you to write a short research report, not fill in an answer sheet. Work through each part in sequence. For every sub-question, follow this structure:

- **State your purpose.** Say in one sentence what you are doing and why, before running any code.
- **Show your code.** Include the code used to produce each result, cleanly formatted.
- **Show key output.** Report results as a table or figure. Do not paste raw console output dumps.
- **Interpret.** Write 2–4 sentences explaining what the result means, what its limitations are, and how it connects to the question of causality.

Interpretation is weighted most heavily. A correct regression table with no interpretation scores poorly. A well-reasoned discussion of an imperfect result scores well. Be concise — one tight paragraph is better than three vague ones.

You may use **R** (.Rmd knitted to PDF) or **Stata** (.do file + Word document). Submit **one file only**. Your file must contain code, output, and interpretation throughout.

Do not copy code, tables, or interpretations from others. Plagiarism is easy to detect. Academic integrity violations will be penalised.

Mark Allocation

Part	Section	Marks
Part 0	Data Familiarisation	6
Part 1	Causal Framework	9
Part 2	Observational Estimation	15
Part 3	Critical Reflection	5
Part 4	Programme Evaluation and IV	15
Total		**50**

Part 4(f) is to be attempted **after Wednesday's lecture on heterogeneous effects** and is worth 5 of the 15 marks in Part 4. All other parts can be completed before that lecture.

Part 0 — Data Familiarisation [6 marks]

(a) Load the dataset. The **income** variable is not recorded in Indian Rupees as stored — make the necessary adjustment. Verify that the corrected income range is economically sensible in the context of the programme.

(b) The dataset does not contain an eligibility variable. Using the programme criteria stated above, construct it yourself. How many households are eligible? What share of the sample does this represent?

[2 marks]

> **Hint**
> *Make sure you have applied the income correction from (a) before constructing eligibility. Check: does your constructed variable agree with your expectations given the income and household distributions?*

(c) Produce a summary statistics table for all key variables, reported separately for net users and non-users. What do you notice? Do net users look systematically different from non-users on observable characteristics? Why does this pattern matter for the analysis that follows?

Additionally, produce at least one plot (e.g. a density plot or box plot) showing the distribution of **dengue_risk** by net use status. Comment briefly on what you observe.

[4 marks]

Part 1 — Causal Framework [9 marks]

(a) State the research question as a precise causal question. What is the estimand you are targeting — the Average Treatment Effect (ATE), the Average Treatment Effect on the Treated (ATT), or the Average Treatment Effect on the Untreated (ATU)? Does your choice of estimand matter here? Justify briefly.

Also consider: does it matter whether treatment is defined as binary (**net**: yes/no) or as a count (**net_num**: 0–3)? Do you expect the effect of the first net to equal the effect of each additional net? What are the implications for your analysis? A

[*NOTE: you have to create the **net** variable. Think about the following points.*

- For your **regression and matching analysis**, which treatment variable is more appropriate, and why?
- For your **IV analysis in Part 4**, which treatment variable makes more sense as the endogenous variable, and why?]

[2 marks]

(b) Draw a Directed Acyclic Graph (DAG) representing the causal structure of this problem. Your DAG must:

- Identify all confounders — variables that cause both net use and dengue risk.
- Identify purely environmental variables — variables that affect dengue risk but do not affect the decision to use nets.
- Show clearly where eligibility fits in the graph.
- Identify any variable you should **not** control for, and explain why.

[4 marks]

> **Hint**
>
> *Think carefully about what actually drives a household's decision to use a net.*
>
> *Ask: does rainfall cause households to use more or fewer nets? Does temperature? Think about what the household observes when making its decision.*
>
> *Ask: does controlling for a variable on the causal pathway from net use to dengue risk make sense?*

(c) Using your DAG, determine the **minimum sufficient adjustment set** — the smallest set of variables you must condition on to identify the causal effect of net use on dengue risk. Justify each variable you include and each variable you choose to exclude.

[3 marks]

Part 2 — Observational Estimation [15 marks]

Work through the steps below in sequence. For each method, state your estimate clearly and compare it to the previous step. A summary comparison table is required at the end.

(a) Naive model. Regress **dengue_risk** on **net** alone. Report and interpret the coefficient. Then repeat the regression using **net_num** as the treatment variable. How do the two estimates compare?

Why are neither of these estimates causal? Use your DAG from Part 1(b) to name the specific sources of bias — do not just say "confounding".

[2 marks]

(b) Regression adjustment. Add the variables from your minimum adjustment set to the regression (do this for both the binary and count treatment specifications). How does the estimated treatment effect change? Interpret the adjusted coefficient.

Note: You may also include the purely environmental variables (**temperature**, **resistance**, **rainfall**) in the regression, even if they are not confounders. Does adding them change the point estimate? Does it affect the standard error? Why?

What assumptions must hold for the adjusted OLS estimate to be interpreted as causal?

[3 marks]

(c) Matching. Use matching to close the backdoors identified in your DAG. You should:

- Implement at least two matching methods, for example: nearest-neighbour matching on covariates, and propensity score matching.
- Report covariate balance before and after matching using a balance table or love plot. Does matching successfully improve balance?
- Report the treatment effect estimate from each method.
- Discuss why different matching procedures may give different estimates, and justify which you would prefer and why.

[5 marks]

> **Hint**
> *R: use the MatchIt package. Stata: use teffects psmatch or teffects nnmatch.*
> *For balance, report standardised mean differences (SMD) before and after matching. A SMD below 0.1 is generally considered acceptable balance.*

(d) Inverse Probability Weighting (IPW). Estimate propensity scores for net use using your adjustment set. Construct IPW weights. Before running the weighted regression, inspect the weight distribution: are there extreme weights? If so, consider trimming and explain your decision. Report and interpret the IPW estimate.

[3 marks]

> **Hint**
> *R: use the WeightIt package (method = "ps").*
> *Stata: use teffects ipw.*
> *Stretch (optional, ungraded): After constructing weights, also include the adjustment-set covariates directly in the weighted regression. This is a doubly robust estimator — it gives a consistent estimate if*

> *either the propensity score model or the outcome model is correctly specified, but not necessarily both. Why is this appealing? [R: lm_weightit() after WeightIt; Stata: teffects ipwra]*

(e) Comparison. Produce a single summary table showing all estimates from Parts 2(a)–(d) side by side, with standard errors. What is the overall pattern? What does the gap between the naive estimate and the adjusted estimates tell you about the direction and magnitude of confounding in this data?

[2 marks]

Part 3 — Critical Reflection [5 marks]

(a) Under what assumptions are the estimates in Part 2 causal? State each assumption in plain language — not just its name. Do you find these assumptions credible in this setting? Be specific about which you are more and less confident in.

[2 marks]

(b) What threats to identification remain even after your adjustments? Think carefully about: unobserved confounders, the possibility that some households select into net use based on factors you cannot observe, and any features of the data that give you pause.

[2 marks]

(c) What additional data, or what alternative research design, would most improve your confidence in a causal claim? Be specific and brief.

[1 mark]

Part 4 — Programme Evaluation and Instrumental Variables [15 marks]

The government programme offers an alternative route to identification. Parts (a)–(e) can be completed now. Part (f) should be attempted after Monday's lecture.

(a) Is evaluating the government programme the same question as estimating the effect of net use in Parts 1–3? Carefully distinguish between the following three estimands in the context of this problem:

- The Intent-to-Treat (ITT) effect of the programme
- The Local Average Treatment Effect (LATE) of net use
- The Average Treatment Effect (ATE) of net use

To whom does each apply? Which one does the IV estimator recover, and why?

[2 marks]

(b) Propose eligibility as an instrumental variable for net use. A valid instrument must satisfy two conditions. State them precisely and assess whether each holds in this setting.

Pay particular attention to the **exclusion restriction**. Does eligibility affect dengue risk through any channel other than net use? Think carefully about what determines eligibility.

[3 marks]

> **Hint**
> *Eligibility is determined by income AND household size. Ask: do either of these directly affect dengue risk?*
> *If they do, eligibility is not a valid unconditional instrument. What do you need to condition on, and why does conditioning on these variables restore validity?*
> *Be honest in your assessment — if the exclusion restriction is only approximately satisfied, say so, and discuss what that implies for your IV estimate.*

(c) Test the first stage: regress **net** on **eligible** (and any controls needed for the exclusion restriction). Report the coefficient and an appropriate test statistic (F-statistic or equivalent). Is the instrument strong or weak? What are the consequences of using a weak instrument?

[2 marks]

(d) Estimate the 2SLS (IV) model. Report and interpret the coefficient on net use. Compare it to your adjusted OLS estimate from Part 2(b). Are they similar? If they differ meaningfully, what are two possible explanations?

[3 marks]

> **Hint**
> *R: use ivreg() from the AER package. Syntax: ivreg(dengue_risk ~ net + controls | eligible + controls, data = df)*
> *Stata: use ivregress 2sls or ivreg2 if installed.*
> *Remember to include the determinants of eligibility as controls on both sides of the | separator.*

(e) The IV estimate recovers a LATE — the average treatment effect for **compliers** only. Who are the compliers in this setting? Characterise them in terms of the eligibility criteria. How large is the complier group relative to the full sample?

Is the LATE likely to be larger, smaller, or similar to the ATE? Give a reasoned answer. What does this imply for the policy relevance of your IV estimate — for whom can you generalise the finding?

[3 marks]

(f) ***To be attempted after Wednesday's lecture***

Explore whether the treatment effect is heterogeneous. Does the effect of mosquito net use on dengue risk differ across households — for example, by income level, household size, or baseline health? Estimate a heterogeneous treatment effect and interpret your findings. What does the presence (or absence) of heterogeneity imply for how the government should design and target the programme?

[2 marks]

All the best.